

HATGAL, Mongolia

WMO: 442070

Lat: 50.4394N

Lon: 100.1508E

Elev: 1668

StdP: 82.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.1	-34.6	-40.1	0.1	-36.9	-37.7	0.1	-34.4	10.4	-15.8	10.0	-14.9	1.5	270	0.801

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	23.7	13.7	21.8	12.8	20.0	12.2	15.1	21.3	14.0	19.8	13.1	18.5	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.7	11.2	17.4	11.6	10.4	16.3	10.6	9.8	15.7	48.3	21.2	44.9	19.8	42.2	18.5	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.9	8.8	8.0	-38.6	27.4	3.4	2.0	-41.1	28.9	-43.1	30.0	-45.0	31.1	-47.5	32.5
			-39.1	17.2	3.3	1.3	-41.4	18.1	-43.4	18.9	-45.2	19.6	-47.6	20.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.3	-22.8	-19.0	-10.5	-1.0	4.8	10.6	13.3	11.7	5.8	-2.3	-12.2	-19.5	(6)
(7)		DBStd	13.46	5.63	6.11	6.38	5.08	4.50	3.26	2.54	2.97	3.92	5.30	6.42	6.24	(7)
(8)		HDD10.0	5107	1017	812	635	331	168	31	3	17	134	381	666	913	(8)
(9)		HDD18.3	7911	1276	1046	893	580	419	231	157	207	377	639	916	1172	(9)
(10)		CDD10.0	241	0	0	0	1	8	50	106	69	7	0	0	0	(10)
(11)		CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)
(12)		CDH23.3	69	0	0	0	0	3	18	30	17	1	0	0	0	(12)
(13)		CDH26.7	7	0	0	0	0	0	1	3	3	0	0	0	0	(13)
(14)	Wind	WSAvg	3.1	2.7	2.9	3.5	3.7	3.6	3.0	2.7	3.1	3.3	3.4	3.1	(14)	
(15)	Precipitation	PrecAvg	288	1	1	3	10	16	48	85	78	33	13	5	2	(15)
(16)		PrecMax	433	9	5	16	60	63	150	199	196	122	64	21	10	(16)
(17)		PrecMin	199	0	0	0	0	0	4	17	17	2	0	0	0	(17)
(18)		PrecStd	67	2	1	3	11	13	26	43	38	21	13	5	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.7	-1.0	9.6	17.2	23.5	25.9	26.8	25.2	21.5	14.2	5.7	-3.4	(19)
(20)			MCWB	-7.6	-4.4	3.0	7.2	11.2	13.6	15.6	15.0	10.9	6.0	0.7	-6.0	(20)
(21)		2%	DB	-9.2	-3.9	5.7	13.4	20.2	22.8	23.7	22.3	18.5	11.2	2.2	-6.2	(21)
(22)			MCWB	-10.8	-6.6	0.3	4.7	9.3	12.5	14.7	13.7	9.8	4.4	-1.4	-8.2	(22)
(23)		5%	DB	-11.8	-6.6	2.8	10.7	17.5	20.6	21.7	20.2	16.1	8.6	-0.4	-8.4	(23)
(24)			MCWB	-13.0	-8.7	-1.6	3.2	8.1	11.7	13.7	12.9	8.8	2.9	-3.2	-10.0	(24)
(25)		10%	DB	-14.1	-9.4	0.0	8.4	14.8	18.6	19.7	18.4	13.7	6.3	-2.8	-10.4	(25)
(26)			MCWB	-15.1	-11.0	-3.4	2.0	6.6	10.9	13.2	12.3	7.6	1.5	-5.1	-11.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.5	-4.2	3.9	7.7	12.2	14.7	17.2	16.0	12.3	6.7	1.1	-5.9	(27)
(28)			MCDWB	-5.6	-1.2	9.5	16.2	22.1	23.3	24.5	23.1	18.8	13.6	5.2	-3.8	(28)
(29)		2%	WB	-10.6	-6.4	0.4	5.2	10.1	13.4	15.6	14.6	10.7	4.9	-1.3	-8.1	(29)
(30)			MCDWB	-9.0	-3.8	5.4	12.6	19.3	20.7	21.7	20.7	17.1	10.8	2.1	-6.2	(30)
(31)		5%	WB	-13.0	-8.5	-1.4	3.6	8.4	12.5	14.6	13.6	9.3	3.4	-3.1	-9.8	(31)
(32)			MCDWB	-11.8	-6.3	2.7	10.2	16.6	19.1	20.1	18.9	14.9	8.7	-0.4	-8.3	(32)
(33)		10%	WB	-15.1	-10.8	-3.2	2.1	6.9	11.5	13.6	12.7	8.0	1.9	-5.0	-11.6	(33)
(34)			MCDWB	-14.1	-9.2	0.2	8.0	14.3	17.6	18.7	17.6	13.1	6.5	-2.7	-10.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.8	15.0	15.5	13.5	14.2	12.8	11.1	11.6	13.4	13.2	11.9	10.5	(35)
(36)			MCDBR	12.8	16.4	16.6	17.5	19.7	17.8	15.8	15.5	17.7	16.1	13.3	11.1	(36)
(37)			MCWBR	11.7	14.0	12.2	10.7	11.2	9.4	8.1	8.3	10.1	10.4	10.2	9.6	(37)
(38)		5% WB	MCDWB	12.9	16.2	16.1	16.6	18.2	14.8	13.2	13.2	15.5	15.7	12.8	10.9	(38)
(39)			MCWBR	11.7	13.9	11.9	10.5	10.9	8.3	7.4	7.4	9.2	10.4	10.1	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.222	0.245	0.287	0.329	0.354	0.349	0.341	0.322	0.286	0.267	0.244	0.225	(40)	
(41)		taud	2.330	2.298	2.255	2.297	2.269	2.395	2.458	2.486	2.530	2.485	2.365	2.296	(41)	
(42)		Ebn at Noon	852	911	920	914	900	907	909	913	920	879	809	783	(42)	
(43)		Edh at Noon	62	85	108	118	127	113	105	97	82	70	60	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.50	2.54	4.04	5.42	6.17	6.19	5.88	5.08	4.17	2.79	1.63	1.16	(44)	
(45)		RadStd	0.06	0.09	0.17	0.24	0.35	0.36	0.31	0.26	0.24	0.15	0.09	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-181	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page